EC242 Group Project 1

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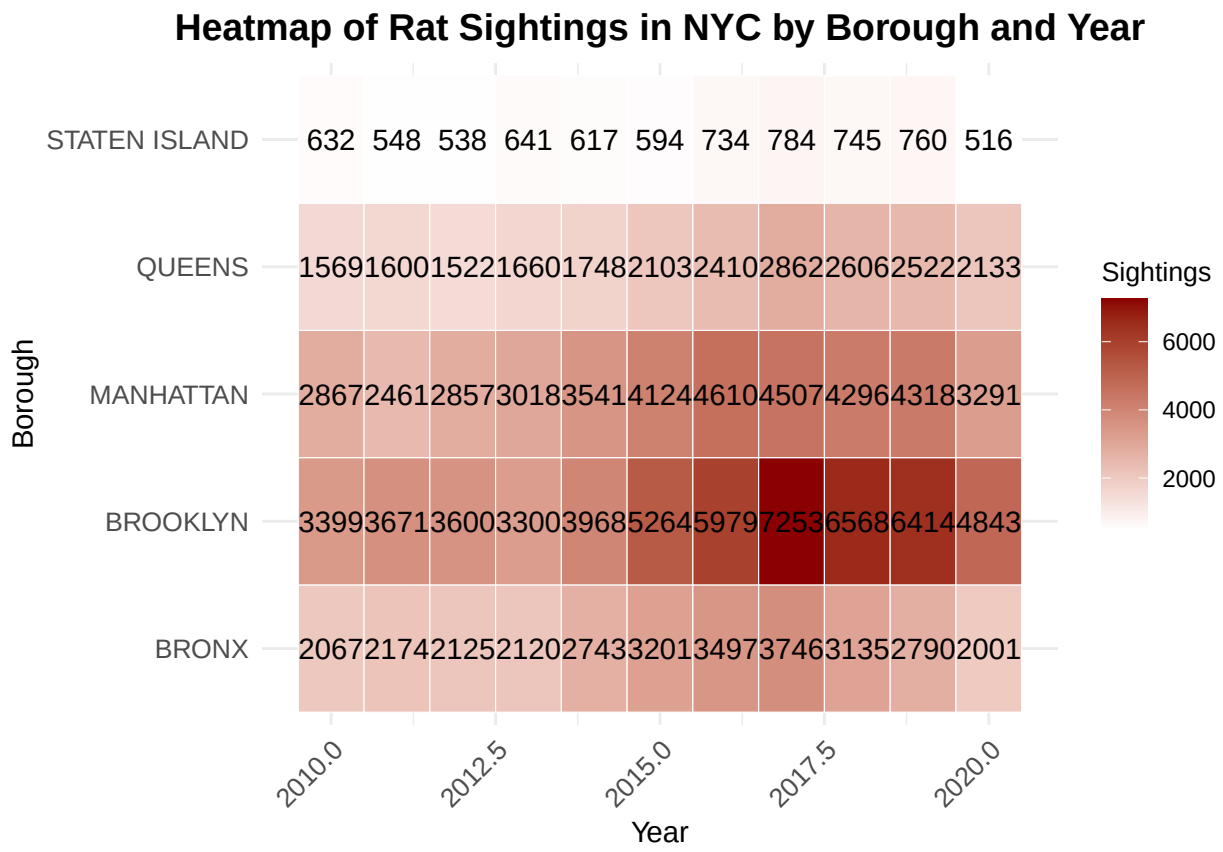
17 September, 2025

Part 1

Rat Sightings Analysis

Data Processing & Cleaning

The dataset was cleaned by standardizing date formats, extracting relevant time-based variables, and filtering out missing borough data to ensure accurate analysis.

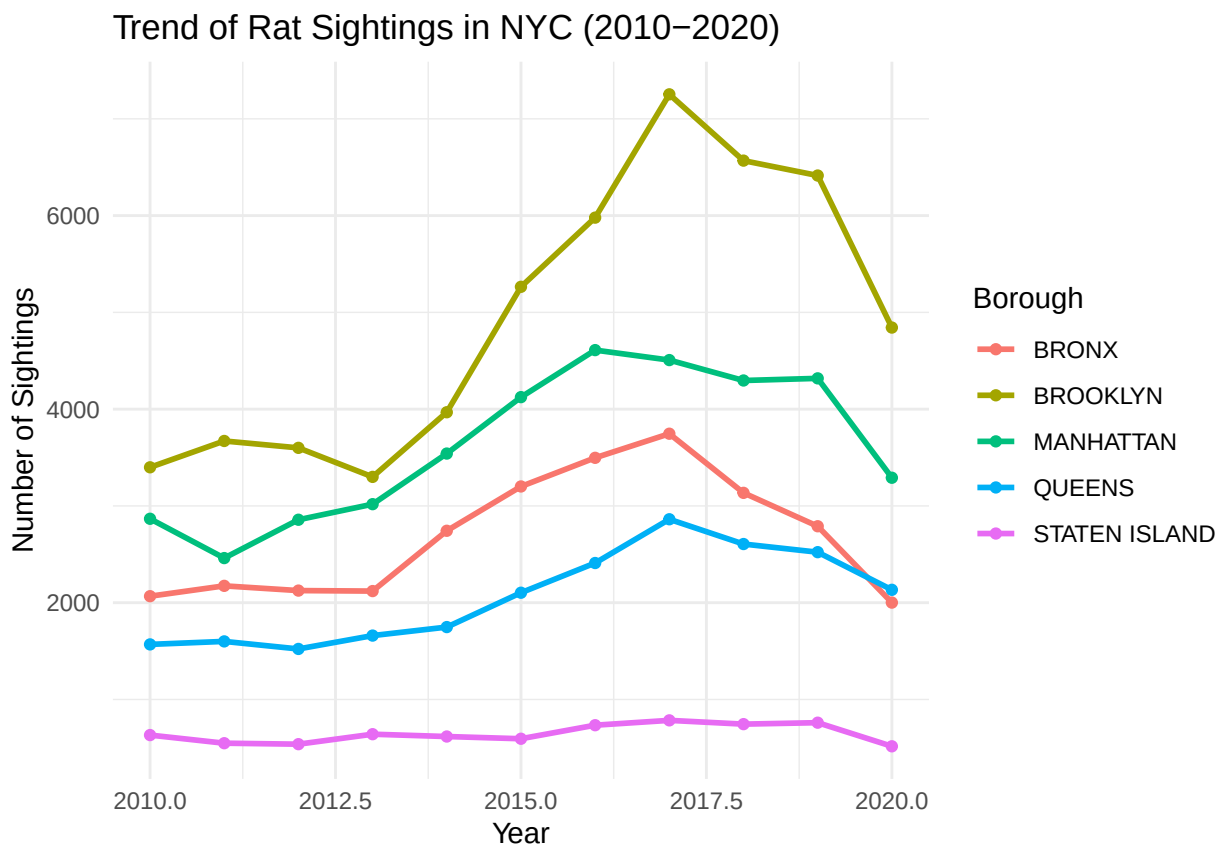


The heatmap clearly shows the following:

- Brooklyn and Manhattan have the most rat sightings.

- There were major increases in rat reports during 2015-2017 throughout several boroughs.
- Queens and Staten Island have fewer rat sightings overall.

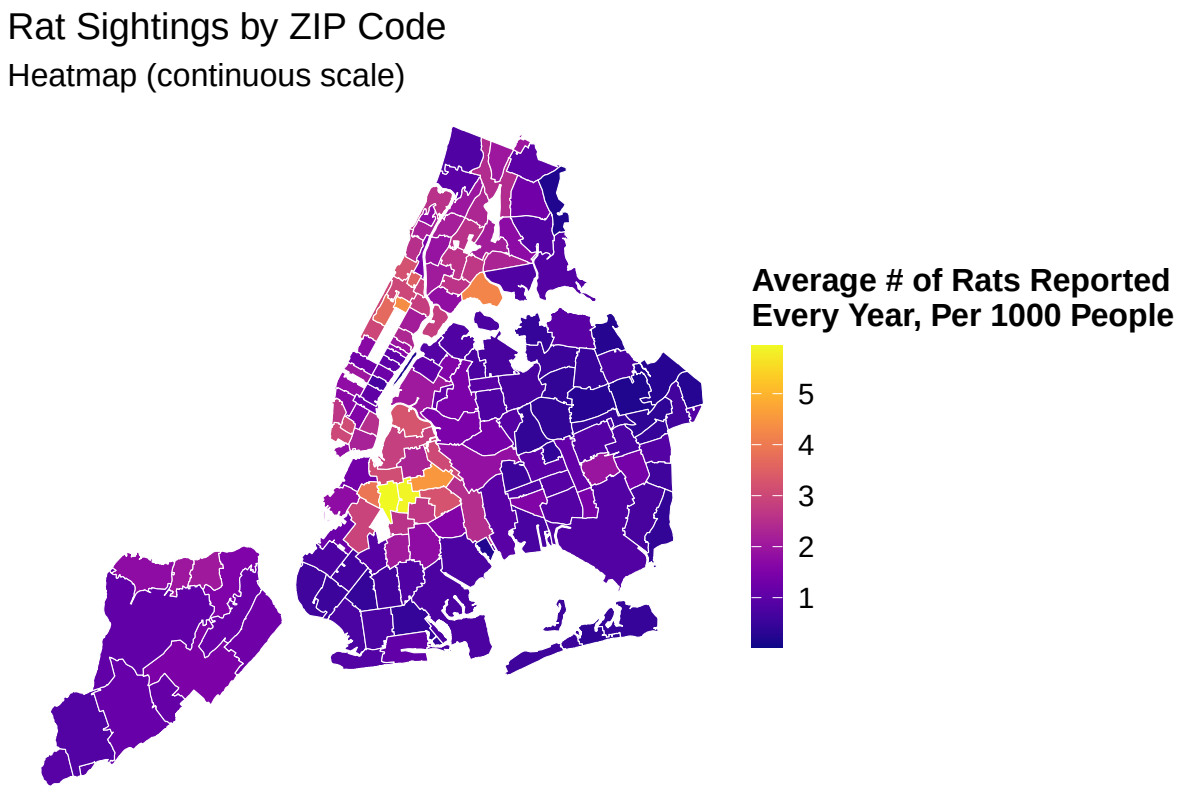
Based on this, we can say that pest control teams should concentrate their efforts in the boroughs with the highest number of rats, mainly Brooklyn and Manhattan.




```
## # A tibble: 193 x 3
##    ZIP   Median_Income Population
##    <chr>         <dbl>      <dbl>
##  1 07002         81285      70468
##  2 07008         87712      25220
##  3 07020        137847      14544
##  4 07024        105594      39785
##  5 07030        176943      58340
##  6 07036         89942      45101
##  7 07047         75505      61202
##  8 07086        122653      16946
##  9 07093         70389      63321
## 10 07202         68071      44638
## # i 183 more rows
```

NYC ZIP Code Boundaries

Source: NYC Open Data

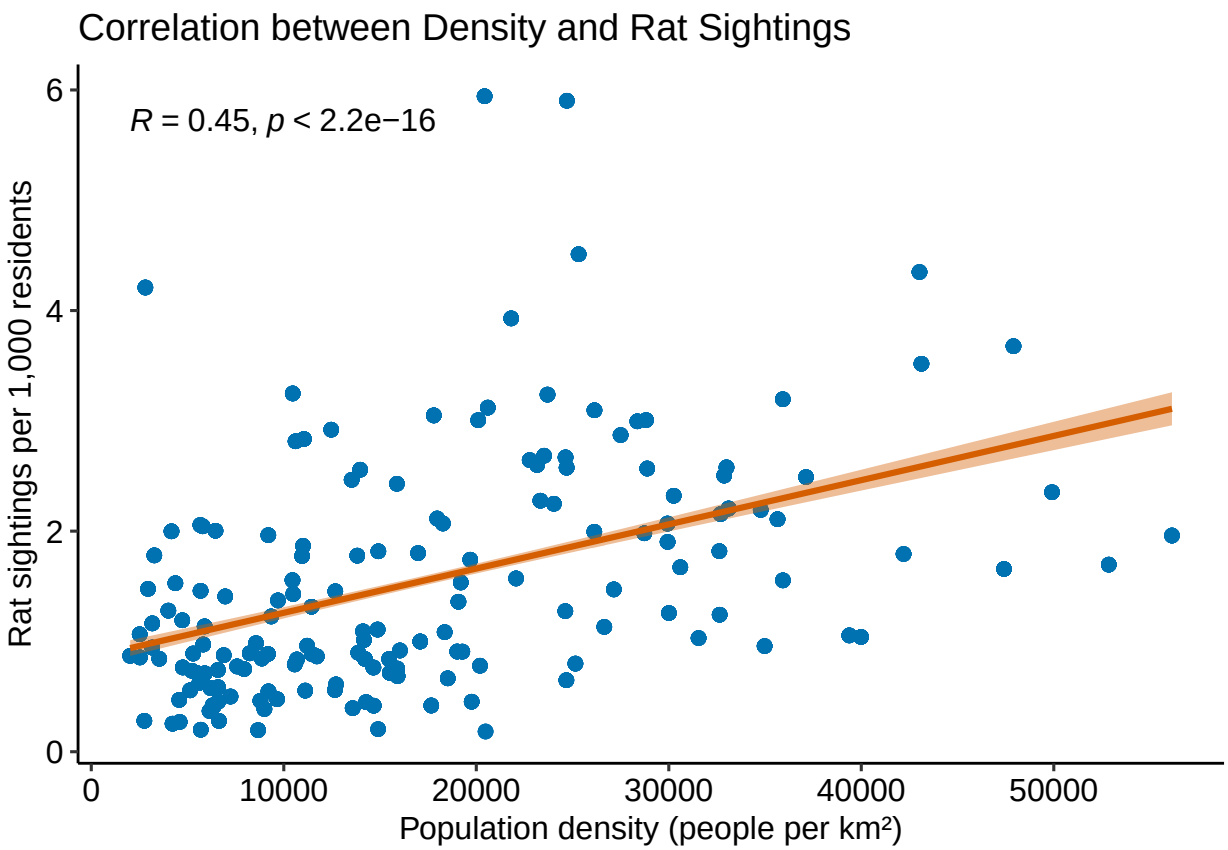


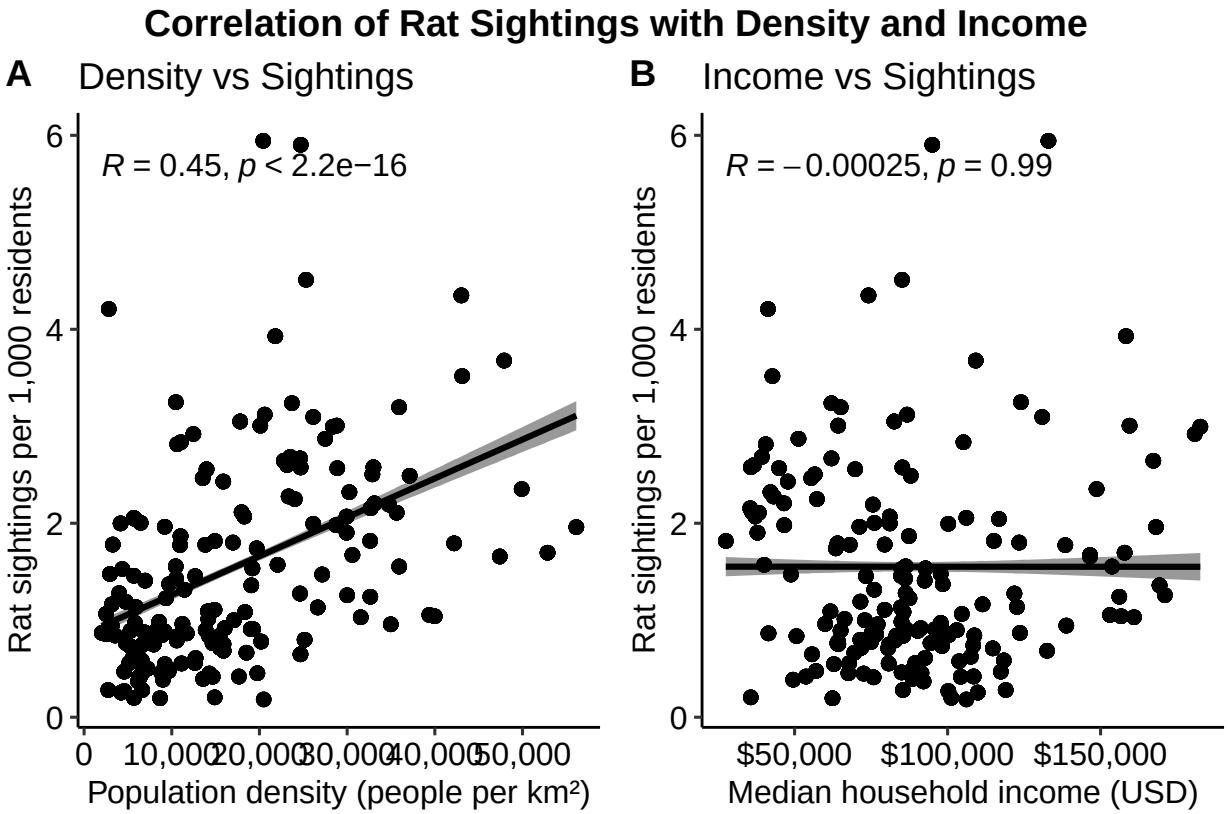
Source: NYC 311 Rat Sightings; Author's calculations

```
##
## Call:
## lm(formula = Sightings_per_1000_per_Year ~ Median_Income + pop_density_sqkm,
##     data = full_dataset_sf)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.4610 -0.6776 -0.2343  0.4289  4.3386
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)       9.720e-01  2.250e-01   4.319 2.72e-05 ***
## Median_Income    -1.408e-06  2.178e-06  -0.647    0.519
## pop_density_sqkm  4.013e-05  6.360e-06   6.310 2.56e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9615 on 162 degrees of freedom
## Multiple R-squared:  0.1973, Adjusted R-squared:  0.1874
## F-statistic: 19.91 on 2 and 162 DF,  p-value: 1.863e-08
```

```
## # A tibble: 6 x 7
##   ZIP    Year borough   Sightings Sightings_per_1000_per_Year pop_density_sqkm
##   <chr> <dbl> <chr>         <int>                       <dbl>            <dbl>
## 1 10001  2010 MANHATTAN        15                        1.80           16977.
```

```
## 2 10001  2011 MANHATTAN        18                        1.80           16977.
## 3 10001  2012 MANHATTAN        28                        1.80           16977.
## 4 10001  2013 MANHATTAN        39                        1.80           16977.
## 5 10001  2014 MANHATTAN        67                        1.80           16977.
## 6 10001  2015 MANHATTAN        60                        1.80           16977.
## # i 1 more variable: Median_Income <dbl>
```




Correlation of Rat Sightings with Density and Income
A
Density vs Sightings
R = 0.45, p < 2.2e−16
Rat sightings per 1,000 residents
0
2
4
6
0
10,000
20,000
30,000
40,000
50,000
Population density (people per km²)
B
Income vs Sightings
R = −0.00025, p = 0.99
Rat sightings per 1,000 residents
0
2
4
6
$50,000
$100,000
$150,000
Median household income (USD)
Points = ZIPs; line = OLS with 95% CI